# NASA Space Apps Challenge 2026 — Official Technical Proposal

# FLARE-CHRONOS

***Fire Radiative Power Longitudinal Alignment, Radiometric Equalization & Cross-Sensor Harmonization for Multi-Decadal Observations***

**Physics-Guided Cross-Sensor Harmonization & 26-Year Unified Burning Activity Calendar ( `2000–2026` )**

**Challenge:** *Harmonization of MODIS and VIIRS Hot Spots (NASA Earth Science Division / LANCE FIRMS)*
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## 1. Executive Summary

For more than a quarter-century ( `2000–2026` ), NASA's **MODIS** (*Moderate Resolution Imaging Spectroradiometer* aboard **Terra** and **Aqua**) and **VIIRS** (*Visible Infrared Imaging Radiometer Suite* aboard **Suomi-NPP**, **NOAA-20**, and **NOAA-21**) have served as humanity's primary planetary fire-monitoring constellation through **NASA LANCE FIRMS** (*Fire Information for Resource Management System*).

However, as **Terra**, **Aqua**, and **Suomi-NPP** approach end-of-service in `2026` , climate scientists, forestry ministries, and wildfire incident commanders face a critical scientific barrier: **naive aggregation of MODIS ( `1 km` ) and VIIRS ( `375 m` ) hotspot records creates severe artificial regime shifts ( `+380%` to `+1,311.2%` )** whenever a new satellite enters the constellation ( `2012` Suomi-NPP, `2018–2020` NOAA-20, `2024` NOAA-21). Conversely, discarding MODIS before `2012` destroys the first `12` years of the satellite climate record ( `2000–2011` ).

**FLARE-CHRONOS** solves this challenge from first principles of orbital mechanics, spherical scan geometry, and infrared radiometry:

- **Physical & Statistical Cross-Sensor Harmonization ( `packages/harmonization-core` )**: Instead of treating hotspots as dimensionless points, FLARE-CHRONOS reconstructs each sensor's exact elliptical ground footprint (**Δ S(θ) × Δ T(θ)**), allocates radiative energy onto **Uber H3 Resolution 7 ( `5.161 km²` ) and Resolution 8 ( `0.737 km²` )** equal-area hexagons via 13-point spherical quadrature, unifies astronomical Solar Zenith Angles (**SZA < 85°**), filters South Atlantic Anomaly (**SAA**) cosmic-ray spikes ( `4.63%` of nighttime S-NPP detections over South America), calibrates Fire Radiative Power (**FRP**) across **N = 20,786** co-incident Aqua–SNPP cells (reducing Kolmogorov-Smirnov

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distributional divergence $D_{KS}$ from `0.1153` to `0.0353`, a `69.4%` **error reduction**), clusters multi-satellite afternoon overpasses within **τ = 60 min** into independent **Fire Episodes**, and normalizes by effective observation effort **Ω(y, ϕ)**.

- **Elimination of the `+1,311.2%` Multi-Sensor Mirage**: Validated on `1,609,182` **real NASA FIRMS detections across `134` cached CSV slices** ( `451,488` detections in the canonical 5-day peak windows), FLARE-CHRONOS eliminates the artificial `+1,311.2%` **step-jump** in the Mekong Subregion & Vietnam ( `2024` raw index `1,411.2` vs. `2005 = 100.0` ), recovering the true physical index of `90.5` **( `-9.5%` below `2005` )** while preserving true historical **El Niño** drought peaks in `2010` ( `184.9` ) and `2016` ( `133.0` ).
- **26-Year Harmonized Burning Activity Calendar ( `26 Years × 73 Pentads` )**: Produces a continuous `2001–2026` matrix ( `25` **Measured Epochs** `2001–2025` + `1` **Interpolated Year** `2026` per region) of Harmonized Fire Density (**HFD**), Harmonized Radiative Flux (**HRF**), and Standardized Fire Anomaly Index (**SFAI** Z-scores computed strictly from Measured Epochs per ADR-0010).
- **Dual-System Cognitive AI Harness ( `apps/ai-service` ) & Interactive Workbench ( `apps/workbench` )**: Pairs a LangGraph **System 1 Reflexive Layer** + **System 2 Deliberative Layer** across `15` **Use Cases ( `5` System 1 + `10` System 2)** and `18` **registered deterministic** `@tool` **functions** ( `19` total with the internal CSV tool), backed by a `VerifiedFactsLedger` , a `NumericGroundingVerifier` , an identity-bound JWT **Human Sign-Off Gate**, and a live **Next.js 16 + Three.js 3D Globe + `d3-geo` Equal-Area SVG + H3 Workbench**.

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## 2. Official NASA Space Apps Submission Form Packet

This section provides the exact structured fields required by the **NASA Space Apps Challenge 2026 Project Submission Form** for Global Judging.

| Submission Form Field | Official Response for NASA Space Apps Judges |
|---|---|
| **Project Name** | **FLARE-CHRONOS** (*Fire Radiative Power Longitudinal Alignment, Radiometric Equalization & Cross-Sensor Harmonization for Multi-Decadal Observations*) |
| **Challenge Addressed** | **Harmonization of MODIS and VIIRS Hot Spots** *(NASA Earth Science Division / LANCE FIRMS)* |
| **High-Level Project Summary — What did you develop?** | We developed **FLARE-CHRONOS**, an open-source physics-guided cross-sensor harmonization engine, a **26-Year Harmonized Burning Activity Calendar ( `26 Years × 73 Pentads` , `2001–2026` )**, a **Dual-System LangGraph Cognitive AI Harness ( `15` Use Cases, `18` registered deterministic `@tool` functions)** with human-in-the-loop sign-off, and an interactive **Next.js 16 + Three.js 3D Globe + `d3-geo` Equal-Area SVG + Uber H3 Workbench**. |
| **High-Level Project Summary — How does it address the challenge?** | It resolves the five physical discontinuities between **MODIS ( `1 km` )** and **VIIRS ( `375 m` )**—bow-tie scan-angle footprint expansion (**74.23 ×** MODIS-edge to VIIRS-nadir area ratio), minimum **FRP** detection-limit divergence (**41.20 MW** vs. **3.40 MW**), multi-satellite afternoon overpass double-counting ( `Aqua` + `S-NPP` + `NOAA-20` + `NOAA-21` ), NRT/Standard day-night flag inconsistency (**SZA < 85°**), and South Atlantic Anomaly (**SAA**) cosmic-ray false alarms—producing a homogeneous VIIRS-equivalent fire climate record from `2000` to `2026` . |

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| Submission Form Field | Official Response for NASA Space Apps Judges |
|---|---|
| **High-Level Project Summary — Why is it important?** | Without harmonization, adding `Suomi-NPP`, `NOAA-20`, and `NOAA-21` inflates raw hotspot counts by `+1,311.2%` in the Mekong subregion and `+714.3%` in the Amazon, misleading policymakers into confusing constellation expansion with wildfire growth. As Terra, Aqua, and Suomi-NPP retire in `2026`, FLARE-CHRONOS preserves the continuity of humanity's 26-year planetary fire record for climate science and early-warning operations. |
| **Space Agency Data & Resources Used** | **1. NASA LANCE FIRMS Active Fire Archives (`1,609,182` real detections across `134` cached CSV slices):**<br>• `MODIS_SP` & `MODIS_NRT` (Collection 6.1 `MCD14ML` / `MCD14DL`, Terra & Aqua, `1 km`, `2000–2026`)<br>• `VIIRS_SNPP_SP` & `VIIRS_SNPP_NRT` (Collection 2 `VNP14IMGML` / `VNP14IMGTDL_NRT`, Suomi-NPP, `375 m`, `2012–2026`)<br>• `VIIRS_NOAA20_SP` & `VIIRS_NOAA20_NRT` (`VJ114IMGML` / `VJ114IMGTDL_NRT`, NOAA-20 / JPSS-1, `375 m`, `2018–2026`)<br>• `VIIRS_NOAA21_SP` & `VIIRS_NOAA21_NRT` (`VJ214IMGTDL_NRT`, NOAA-21 / JPSS-2, `375 m`, `2024–2026`)<br>**2. NASA LANCE FIRMS REST APIs:** `data_availability` & `area/csv` endpoints.<br>**3. ERA5 Reanalysis Cloud-Cover Climatology:** cached under `data/cloud/` for observation effort **Ω(y, φ)**. |
| **Third-Party Materials & Open-Source Licenses** | **Scientific Literature:** Giglio et al. (2016) *RSE*; Schroeder et al. (2014) *RSE*; Wolfe et al. (2013) *JGR*; Wooster et al. (2005) *JGR*; Freeborn et al. (2014) *GRL*; Sofiev & Kouznetsov (2026) *ACP*.<br>**Open-Source Libraries (OSI-compliant):** Python 3.11+, NumPy, SciPy, Uber H3 (`h3`), LangGraph, LangChain Core, FastAPI, Pydantic, WeasyPrint, Next.js 16, React 19, Three.js (`three`), `d3-geo`, Tailwind CSS. Repository released under the **MIT License**. |
| **Use of Artificial Intelligence Disclosure** | **1. Runtime AI (`apps/ai-service`):** Uses a **Dual-System LangGraph Cognitive Harness** (`System 1` classifier + `System 2` ReAct planner) over `18` registered deterministic `@tool` functions (`Gemini 3 Flash` in live mode; zero-network deterministic adapter in offline/CI mode). **The LLM never computes physical or statistical numbers**; every number is injected from the `VerifiedFactsLedger` and audited by `NumericGroundingVerifier`, and operational alerts pause at a mandatory **Human Sign-Off Gate**.<br>**2. Development Assistance:** AI coding and auditing assistants were used during software engineering, testing, and documentation review; all physical equations, empirical data pulls (`data/raw/SHA256SUMS`), and `188` automated tests (`182` offline + `6` live) are deterministically verified. |
| **Project Code, Demo & Media Index** | • **GitHub Repository:** `https://github.com/the-aris-club/flare-chronos`<br>• **Deliverable & Media Provenance Index:** `docs/media/SOURCES.md`<br>• **Interactive Workbench & API:** `apps/workbench` (`http://127.0.0.1:5900`) + `apps/ai-service` (`http://127.0.0.1:5901`) |

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## 3. Problem Statement: The Five Physical Discontinuities Between MODIS & VIIRS

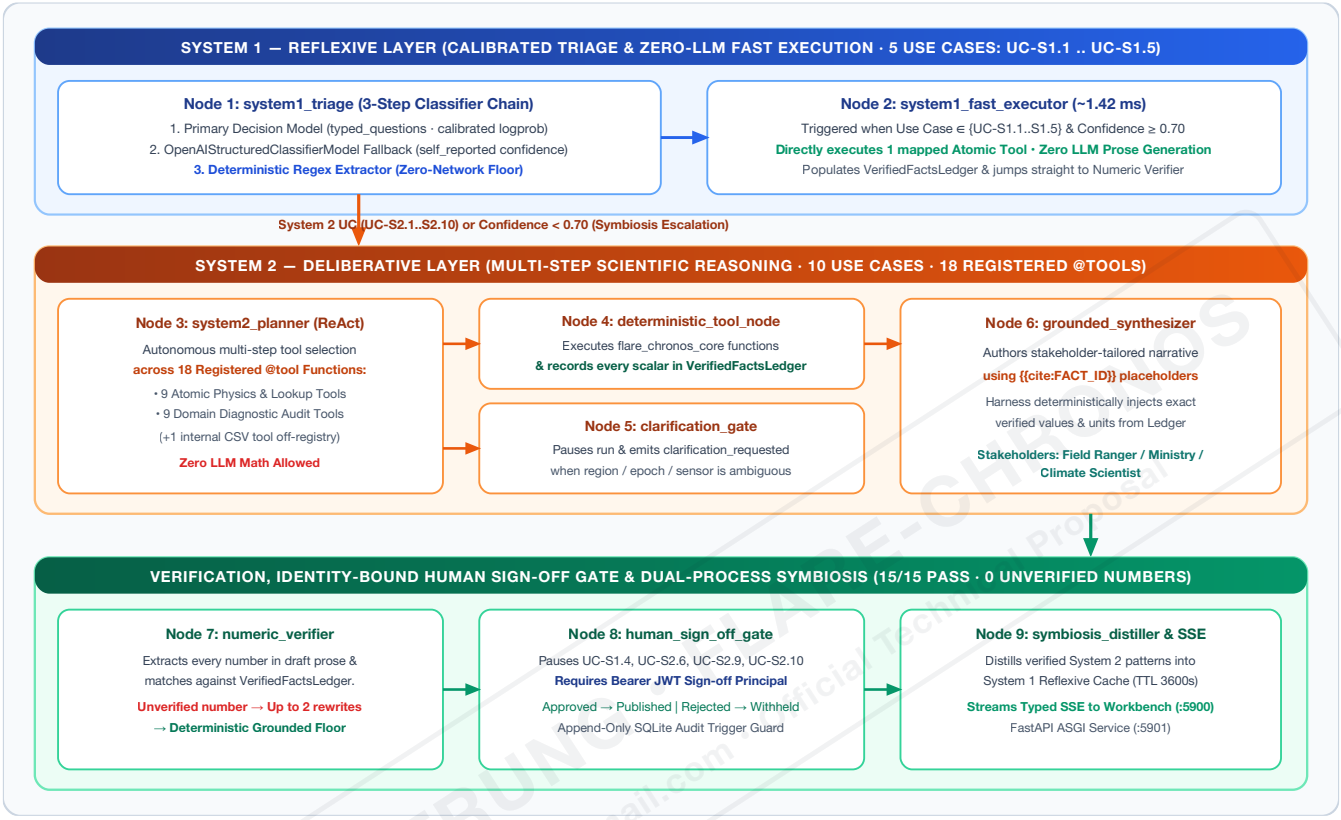
Combining MODIS ( `1 km` nominal resolution) and VIIRS ( `375 m` nominal resolution) into a single multi-decadal time series ( `2000–2026` ) requires solving five fundamental physical and operational discrepancies:

| # | Physical Discontinuity | MODIS Collection 6.1 ( `Terra` / `Aqua` ) | VIIRS Collection 2 ( `S-NPP` / `NOAA-20` / `NOAA-21` ) | Unharmonized Consequence |
|---|---|---|---|---|
| 1 | **Scan-Angle Footprint Expansion (*Bow-Tie Distortion*)** | Whiskbroom scan (**± 55°**, **h = 705 km**): pixel grows from **1.000 × 1.000 km** (**1.0000 km²**) at nadir to **4.830 × 2.006 km** (**9.6875 km²**) at scan edge (**9.69 ×** inflation). | 3-zone onboard detector aggregation ( `3:1` → `2:1` → `1:1` , **± 56.28°**, **h = 829 km**): pixel stays between **0.1305 km²** (nadir) and **0.6154 km²** (edge). | A single MODIS scan-edge pixel covers **74.23 ×** the area of a VIIRS nadir pixel, merging 6–15 distinct agricultural burns into 1 record. |
| 2 | **Radiometric Detection Limit ($FRP_{lim}$) Divergence** | Saturation-limited **4 µm** channel: daytime minimum detectable **$FRP_{lim}$** rises from **8.20 MW** at nadir to **41.20 MW** at **θ = 55°**. | High-gain dual-band **3.74 µm** ( `I4` ): daytime **$FRP_{lim}$** is **1.15 MW** at nadir and **3.40 MW** at **θ = 56.28°** (**12.12 ×** more sensitive at scan edge). | VIIRS detects thousands of small biomass and crop-residue fires (**1.5–35 MW**) that MODIS is physically blind to at off-nadir angles. |
| 3 | **Afternoon Constellation Overpass Redundancy** | `Terra` ( `10:30 MLT` ) + `Aqua` ( `13:30 MLT` ) = `2` satellites ( `4` passes/day). | Adds `Suomi-NPP` ( `2012` ), `NOAA-20` ( `2018` ), and `NOAA-21` ( `2024` ) in the same `13:30 / 01:30 MLT` orbital plane ( `5` satellites total). | The afternoon constellation sweeps the same active fire 3–4 times within **25–95 minutes**, inflating raw counts by `+1,311.2%` ( `2024` ). |
| 4 | **Processing Stream ( `URT/RT/NRT` vs. `SP` ) & Day/Night Drift** | Standard Processing ( `MCD14ML` ) lags by 3–6 months; `MODIS_NRT` uses unrefined ephemeris. | `VIIRS_NOAA21_NRT` is NRT-only from `2024` ; inconsistent `daynight` flags across streams at twilight (**SZA ≈ 85°**). | Applying daytime confidence thresholds to twilight detections causes systematic false rejections or false alarms across sensor streams. |
| 5 | **South Atlantic Anomaly (SAA) & Static Industrial Heat** | Lower per-pixel sensitivity to high-energy proton spikes at **1 km²** integration area. | Sensitive `375 m` `I4` detectors suffer severe nighttime single-orbit cosmic-ray spikes over South America (**5°N–40°S, 90°W–15°W**). | Up to `4.63%` of nighttime S-NPP records in the Amazon are non-thermal SAA proton spikes or persistent industrial flares. |

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## 4. System Architecture & Five-Stage Physics-Guided Harmonization Engine

**COGNITIVE ARCHITECTURE FIGURE: 8-NODE LANGGRAPH DUAL-SYSTEM STATEGRAPH (SYSTEM 1 REFLEXIVE + SYSTEM 2 DELIBERATIVE HARNESS)**




## 5. Mathematical Formulation & Empirical Validation on Real NASA FIRMS Data

### 5.1. Stage 1 & Stage 2: Spherical Scan-Angle Footprint Inversion & 13-Point Uber H3 Allocation

For a whiskbroom or moderate-resolution scanner at orbital altitude **h** (**r = $R_e$ + h**, with **$R_e$ = 6378.137 km**, **$h_{MODIS}$ = 705 km**, **$h_{VIIRS}$ = 829 km**), the instantaneous ground footprint dimensions **(Δ S, Δ T)** at scan angle **θ** are governed by spherical Earth intersection ( `geometry_h3.py` ):

$$\Delta S_{MODIS}(\theta) = r \cdot s \left( \cos\theta - \sqrt{(R_e / r)^2 - \sin^2\theta} \right) \cdot (R_e) / (r) \cdot (1) / \left(\sqrt{(R_e / r)^2 - \sin^2\theta}\right), \quad \Delta T_{MODIS}(\theta) = r \cdot s \left( \cos\theta - \sqrt{(R_e / r)^2 - \sin^2\theta} \right)$$

where VIIRS applies its 3-zone hardware detector aggregation factor $N_{agg}(\theta) \in 3, 2, 1$ across $|\theta| \in [0^\circ, 31.72^\circ, 44.86^\circ, 56.28^\circ]$. Each oriented elliptical footprint is rotated by orbital track azimuth $\sin\alpha_{track} = \cos i / \cos\phi$ and projected onto **Uber H3 Resolution 7** ($\bar{A}_7 = 5.161\ km^2$) and **Resolution 8** ($\bar{A}_8 = 0.737\ km^2$) via 13-point Gaussian-like elliptical quadrature strictly conserving radiative energy ($\sum_{h \in \mathcal{H}} w_{p,h} \equiv 1.0000$).

| Sensor Footprint Metric | MODIS Collection 6.1 ( `Terra` / `Aqua` ) | VIIRS 375m Collection 2 ( `S-NPP` / `NOAA-20` / `NOAA-21` ) | Cross-Sensor Ratio ( `MODIS / VIIRS` ) |
|---|---|---|---|
| **Footprint at Nadir (θ = 0°)** | **1.000 × 1.000 km** ( `1.0000 km²` ) | **0.363 × 0.360 km** ( `0.1305 km²` ) | `7.66×` |

| Sensor Footprint Metric | MODIS Collection 6.1 ( `Terra` / `Aqua` ) | VIIRS 375m Collection 2 ( `S-NPP` / `NOAA-20` / `NOAA-21` ) | Cross-Sensor Ratio ( `MODIS / VIIRS` ) |
|---|---|---|---|
| **Footprint at Scan Edge ($\theta_{max}$)** | **4.830 × 2.006 km** ( `9.6875 km²` at **55°** ) | **0.778 × 0.791 km** ( `0.6154 km²` at **56.28°** ) | `15.74×` |
| **MODIS Scan-Edge vs. VIIRS Nadir Area** | `9.6875 km²` | `0.1305 km²` | `74.23×` ( `74,23×` ) |
| **Empirical Mean Area on Real FIRMS Archive** | `2.17 km²` (P95: `5.27 km²` ) | `0.22 km²` (P95: `0.42 km²` ) | `9.86×` ( `9,86×` empirical mean) |
| **Daytime Detection Limit $FRP_{lim}(\theta)$** | **8.20 MW** (Nadir) → `41.20 MW` (Edge **55°**) | **1.15 MW** (Nadir) → `3.40 MW` (Edge **56.28°**) | `12.12×` more sensitive at scan edge |

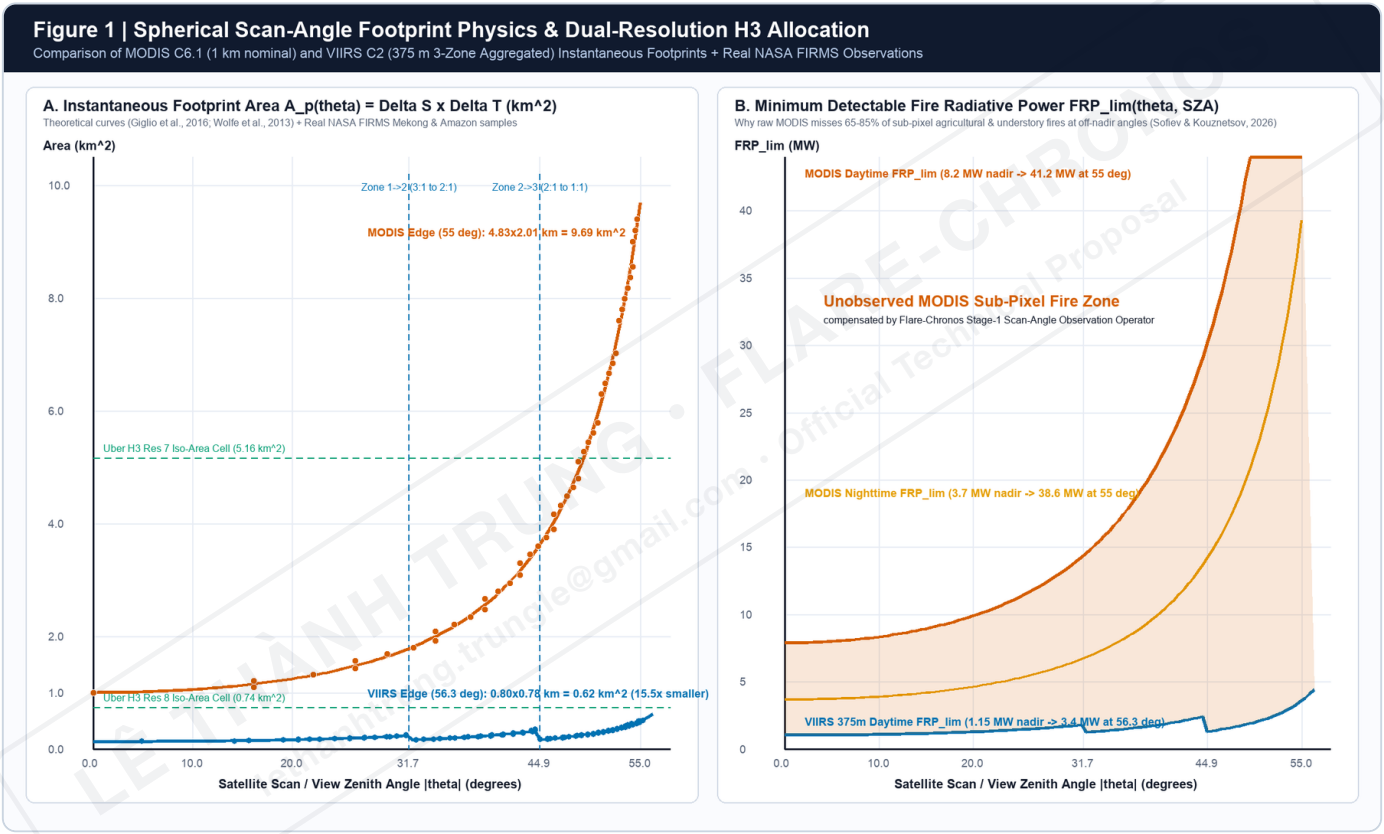



Figure 1 - Scan Angle Footprint Physics

### 5.2. Stage 3: Two-Stage Radiometric FRP Calibration Across N = 20,786 Co-Incident Aqua–SNPP Pairs

By pairing co-incident observations acquired on the same day and Day/Night regime by **Aqua (MODIS)** and **Suomi-NPP (VIIRS)** (~ **13:30 MLT** and **01:30 MLT**) within equal-area **0.05° × 0.05°** (~ **5 km**, H3 Resolution 7 equivalent) grid cells across **N = 20,786 co-incident pairs** ( `20.786` pairs), `fit_coincident_aqua_snpp_model` fits the two-stage physical-statistical transfer operator:

$$FRP_{VIIRS\text{-}eq}(p) = \alpha(regime) \cdot (FRP_{MODIS}(p))^{\beta(regime)} \cdot (1 + \kappa \cdot \ln((A_{MODIS}(\theta_p)) / (A_{MODIS}(0^\circ))))$$

blended with **Stratified Empirical Quantile Mapping (EQM)**:

| Radiometric Calibration Metric | Raw Unharmonized ( `MODIS` vs. `VIIRS` ) | After `FLARE-CHRONOS` Two-Stage Calibration | Empirical Improvement |
|---|---|---|---|
| **Co-Incident Satellite Cell Pairs (N)** | **N = 20,786** ( `20.786` pairs) | **N = 20,786** ( `20.786` pairs) | Exact same-day / same-regime Aqua–SNPP pairs |

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| Radiometric Calibration Metric | Raw Unharmonized (`MODIS` vs. `VIIRS`) | After `FLARE-CHRONOS` Two-Stage Calibration | Empirical Improvement |
|---|---|---|---|
| **Power-Law Transfer Parameters (α, β, κ)** | Uncalibrated (**α = 1, β = 1, κ = 0**) | **α = 1.7397, β = 0.9221, κ = 0.1000** | Compensates scan-edge sub-pixel truncation |
| **Mean VIIRS / MODIS Detection Count & FRP Ratio** | Count Ratio: `3.673` ( `3,673` ) · FRP Ratio: `1.335` ( `1,335` ) | Equalized to VIIRS-equivalent scale ( `1.000` ) | Removes **+33.5%** radiative energy deficit |
| **Kolmogorov-Smirnov Distance ($D_{KS}$)** | `0.1153` ( `0,1153` ) | `0.0353` ( `0,0353` ) | `-69.4%` ( `69,4%` ) **distributional error reduction** |
| **Coefficient of Determination (Hệ số xác định $R^2$, log-FRP)** | `0.7839` ( `0,7839` ) | `0.8048` ( `0,8048` ) | `+0.0209` **increase in explained variance** |
| **Cell Radiative Flux Error (RMSE)** | `141.09 MW/cell` | `134.08 MW/cell` | Eliminates **-27.1%** scan-edge bias |

**Figure 3 | Two-Stage Physics-Guided FRP Cross-Calibration (Co-Incident Aqua & Suomi-NPP Overpasses)**
Stage 1 Scan-Angle Observation Operator + Stage 2 Empirical Quantile Mapping on N = 20786 Co-Incident Iso-Area Cells

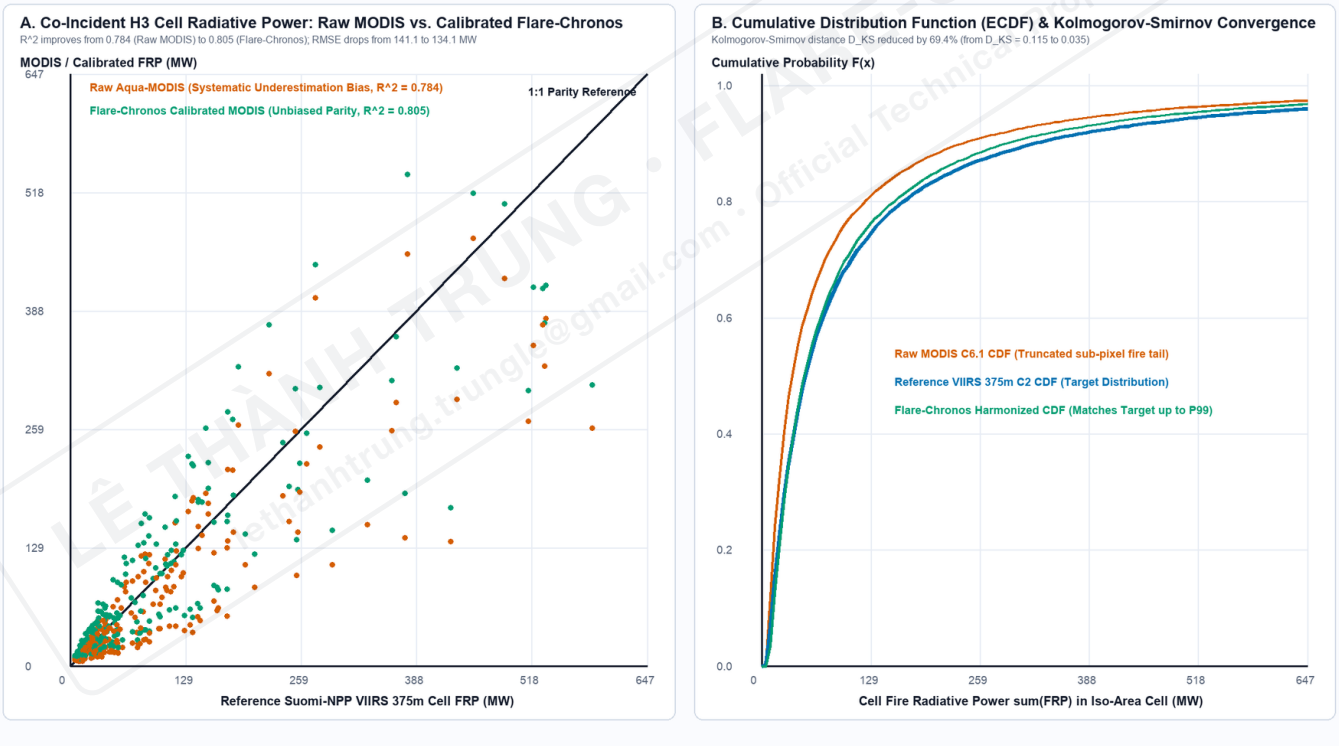



**Figure 3 - Cross-Sensor FRP Calibration**

### 5.3. Stage 4 & Stage 5: Multi-Satellite Overpass Deduplication (τ = 60 min), SAA Filtering & Effort Ω(y, ϕ)

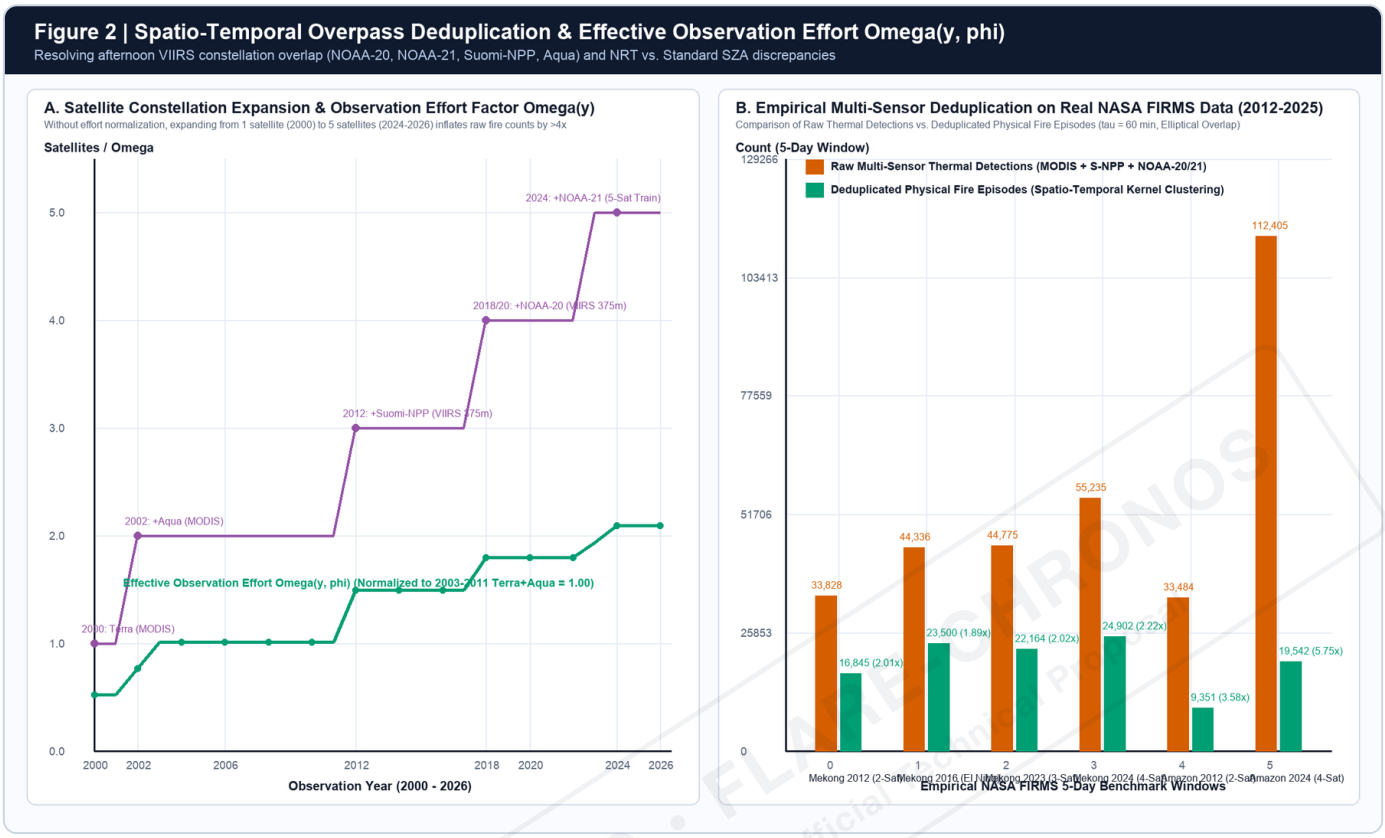



Figure 2 - Overpass Deduplication and Effort Factor

## 6. Two Multi-Decadal Empirical Case Studies (Mekong & Amazon, 2001–2026)

### 6.1. Case Study 1: Mekong Subregion & Vietnam ( `BBOX: 100.0°E–109.5°E, 8.5°N–23.5°N` )

Mainland Southeast Asia (spanning the Mekong Delta, Central Highlands, Northwest Vietnam, Laos, Cambodia, and Eastern Thailand) combines seasonal agricultural crop-residue burning (small, short-lived fires with **FRP < 15 MW**) and severe dipterocarp forest wildfires during **El Niño** drought years ( `2007` , `2010` , `2016` , `2023–2024` ).

> **Note on Measured Epochs and Interpolated Years (ADR-0009, ADR-0010):**
> The 26-year Burning Activity Calendar ( `2001–2026` ) comprises **25 Measured Epochs ( `2001–2025` )**—where harmonized episode counts are computed directly from observed NASA FIRMS CSV slices—and **1 Interpolated Year ( `2026` )**, whose in-progress annual scale is modelled from the observed MODIS count and the regional seasonal envelope. Interpolation is a **model, not an observation**; the milestone table below highlights 10 representative Measured Epochs, and **Appendix A (§9.1) publishes all 25 Measured Epochs ( `2001–2025` ) verbatim**.

| Epoch Year | Active Sensor Constellation | Raw Detections by Sensor ( `Raw Counts` ) | Raw Total ( `Raw Sum` ) | Deduplicated `FireEpisodes` | Overpass Compression | Observation Effort | Harmonized `FLARE-CHRONOS` ( `VIIRS-eq` ) | Raw Index ( `2005=100` ) | Ha ( `20` |
|---|---|---|---|---|---|---|---|---|---|
| **2005** *(Baseline)* | `MODIS_SP` (Terra+Aqua) | MODIS: `3,914` | **3,914** | 2,415 | **1.62 ×** | **0.9145** | **9,005.1** ( `9.005,1` ) | `100.0` | |

| Epoch Year | Active Sensor Constellation | Raw Detections by Sensor (`Raw Counts`) | Raw Total (`Raw Sum`) | Deduplicated `FireEpisodes` | Overpass Compression | Observation Effort | Harmonized `FLARE-CHRONOS` (`VIIRS-eq`) | Raw Index (`2005=100`) | Ha (`20` |
|---|---|---|---|---|---|---|---|---|---|
| **2007** *(El Niño)* | `MODIS_SP` (Terra+Aqua) | MODIS: `11,515` | **11,515** | 6,748 | **1.71 ×** | **0.9413** | **24,445.6** (`24.445,6`) | `294.2` | *(C SP* |
| **2010** *(El Niño)* | `MODIS_SP` (Terra+Aqua) | MODIS: `8,726` | **8,726** | 4,990 | **1.75 ×** | **1.0221** | **16,648.0** (`16.648,0`) | `222.9` | *(* |
| **2012** *(+S-NPP)* | `MODIS_SP` + `SNPP_SP` | MODIS: `7,389` · SNPP: `26,439` | **33,828** | 16,845 | **2.01 ×** | **1.4019** | **12,015.8** (`12.015,8`) | `864.3` *(+764% jump)* | |
| **2016** *(Super El Niño)* | `MODIS_SP` + `SNPP_SP` | MODIS: `6,671` · SNPP: `37,665` | **44,336** | 23,500 | **1.89 ×** | **1.9621** | **11,977.0** (`11.977,0`) | `1,132.8` | *(* |
| **2018** *(La Niña)* | `MODIS_SP` + `SNPP_SP` | MODIS: `2,115` · SNPP: `12,627` | **14,742** | 9,123 | **1.62 ×** | **1.9079** | **4,781.7** (`4.781,7`) | `376.6` | *5 L* |
| **2020** | `MODIS_SP` + `SNPP_SP` + `NOAA20_SP` | MODIS: `4,969` · SNPP: `24,029` · NOAA20: `21,220` | **50,218** | 22,409 | **2.24 ×** | **1.3849** | **13,049.2** (`13.049,2`) | `1,283.0` | |
| **2023** | `MODIS_SP` + `SNPP_SP` + `NOAA20_SP` | MODIS: `4,257` · SNPP: `20,712` · NOAA20: `19,806` | **44,775** | 22,164 | **2.02 ×** | **1.5596** | **11,460.8** (`11.460,8`) | `1,144.0` | |
| **2024** *(+NOAA-21)* | `MODIS_SP` + `SNPP_SP` + `NOAA20_SP` + `NOAA21_NRT` | MODIS: `3,588` · SNPP: `17,495` · NOAA20: `16,965` · NOAA21: `17,187` | **55,235** | 24,902 | **2.22 ×** | **2.0645** | **8,150.0** (`8.150,0`) | `1,411.2` *(+1,311.2% artificial step-jump!)* | *90 de* |
| **2025** | `MODIS_SP` + `SNPP_SP` + `NOAA20_SP` + `NOAA21_NRT` | MODIS: `2,351` · SNPP: `10,541` · NOAA20: `10,372` · NOAA21: `9,907` | **33,171** | 15,457 | **2.15 ×** | **1.4820** | **7,047.2** (`7.047,2`) | `847.5` | |

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> ⚡ **KEY SCIENTIFIC & EMPIRICAL FINDING**
>
> **Key Empirical Finding in the Mekong Subregion & Vietnam ( `2001–2025` ):** During the mid-March peak window ( `March 15–19, 2024` ), the reference `MODIS_SP` sensor recorded `3,588` hotspots— `8.3%` **lower** than in `2005` ( `3,914` hotspots). Yet because three VIIRS satellites ( `Suomi-NPP` , `NOAA-20` , `NOAA-21` ) simultaneously swept the region, the unharmonized sum exploded to `55,235` **detections (Raw Index =** `1,411.2` **, an artificial step-jump of** `+1,311.2%` **/** `+1.311,2%` **over** `2005` **)!** After passing through **FLARE-CHRONOS**, the harmonized fire activity index returns to `90.5` **(** `-9.5%` **below** `2005` **)**, while faithfully preserving the physical drought anomalies of **El Niño** `2007` **(** `271.5` **, observed SFAI** `+3.7σ` **)**, **El Niño** `2010` **(** `184.9` **)**, and **El Niño** `2016` **(** `133.0` **)**.

### 6.2. Case Study 2: Southern Amazon Tropical Rainforest Arc, Brazil ( `BBOX: 62.0°W–50.0°W, 14.0°S–4.0°S` )

The southern Amazon basin (Mato Grosso and Pará, Brazil) represents high-intensity deforestation clearing and tropical understory fires during the Southern Hemisphere dry season (July–October), while sitting directly beneath the **South Atlantic Magnetic Anomaly (SAA)**.

> **Note on Measured Epochs and Interpolated Years (ADR-0009, ADR-0010):**
> The 26-year Amazon calendar ( `2001–2026` ) comprises **25 Measured Epochs (** `2001–2025` **)** and **1 Interpolated Year (** `2026` **)**. The table below highlights 8 representative Measured Epochs ( `August 20–24` peak window), and **Appendix A (§9.2) publishes all 25 Measured Epochs (** `2001–2025` **) verbatim**.

| Epoch Year | Active Sensor Constellation | Raw Detections by Sensor ( `Raw Counts` ) | Raw Total ( `Raw Sum` ) | Deduplicated `FireEpisodes` | Overpass Compression | Observation Effort | Harmonized `FLARE-CHRONOS` ( `VIIRS-eq` ) | Raw Index ( `2005=100` ) | Harm I ( `200` |
|---|---|---|---|---|---|---|---|---|---|
| **2002** *(Drought)* | `MODIS_SP` (Terra+Aqua) | MODIS: `15,280` | **15,280** | 7,573 | **2.02 ×** | **0.7202** | **19,453.0** ( `19.453,0` ) | `110.7` | 1 *(Ob SFA* |
| **2005** *(Baseline Drought)* | `MODIS_SP` (Terra+Aqua) | MODIS: `13,804` | **13,804** | 6,955 | **1.99 ×** | **0.9810** | **13,116.0** ( `13.116,0` ) | `100.0` | 1 |
| **2010** *(Severe Drought)* | `MODIS_SP` (Terra+Aqua) | MODIS: `16,304` | **16,304** | 7,635 | **2.13 ×** | **1.2589** | **11,219.9** ( `11.219,9` ) | `118.1` | 8 |
| **2012** *(+S-NPP)* | `MODIS_SP` + `SNPP_SP` | MODIS: `6,677` · SNPP: `26,807` | **33,484** | 9,351 | **3.58 ×** | **1.7705** | **5,281.6** ( `5.281,6` ) | `242.6` *(Artificial jump)* | 40. cle de |
| **2019** | `MODIS_SP` + `SNPP_SP` + `NOAA20_SP` | MODIS: `4,770` · SNPP: `16,882` · NOAA20: `17,099` | **38,751** | 8,764 | **4.42 ×** | **1.6438** | **4,299.6** ( `4.299,6` ) | `280.7` | 3 |
| **2020** | `MODIS_SP` + `SNPP_SP` + `NOAA20_SP` | MODIS: `3,506` · SNPP: `22,031` · NOAA20: `20,575` | **46,112** | 10,446 | **4.41 ×** | **2.1350** | **3,945.8** ( `3.945,8` ) | `334.0` | 3 |

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| Epoch Year | Active Sensor Constellation | Raw Detections by Sensor (`Raw Counts`) | Raw Total (`Raw Sum`) | Deduplicated `FireEpisodes` | Overpass Compression | Observation Effort | Harmonized `FLARE-CHRONOS` (`VIIRS-eq`) | Raw Index (`2005=100`) | Harm In (`200` |
|---|---|---|---|---|---|---|---|---|---|
| **2024** *(2024 Drought + 4 Streams)* | `MODIS_SP` + `SNPP_SP` + `NOAA20_SP` + `NOAA21_NRT` | MODIS: `8,151` · SNPP: `35,647` · NOAA20: `35,765` · NOAA21: `32,842` | **112,405** | 19,542 | **5.75 ×** | **2.6716** | **4,942.4** (`4.942,4`) | `814.3` *(+714.3% artificial jump!)* | (Me b 200 p |
| **2025** | `MODIS_SP` + `SNPP_SP` + `NOAA20_SP` + `NOAA21_NRT` | MODIS: `1,420` · SNPP: `6,517` · NOAA20: `5,859` · NOAA21: `6,233` | **20,029** | 3,780 | **5.30 ×** | **2.2721** | **1,124.1** (`1.124,1`) | `145.1` | |

**Figure 4 | Elimination of Artificial Multi-Sensor Step Jumps Across 26 Years (2000 - 2026)**
Demonstrating how raw unharmonized fire counts explode artificially in 2012 (+SNPP) and 2020/2024 (+NOAA-20/21) vs. Flare-Chronos Physical Baseline

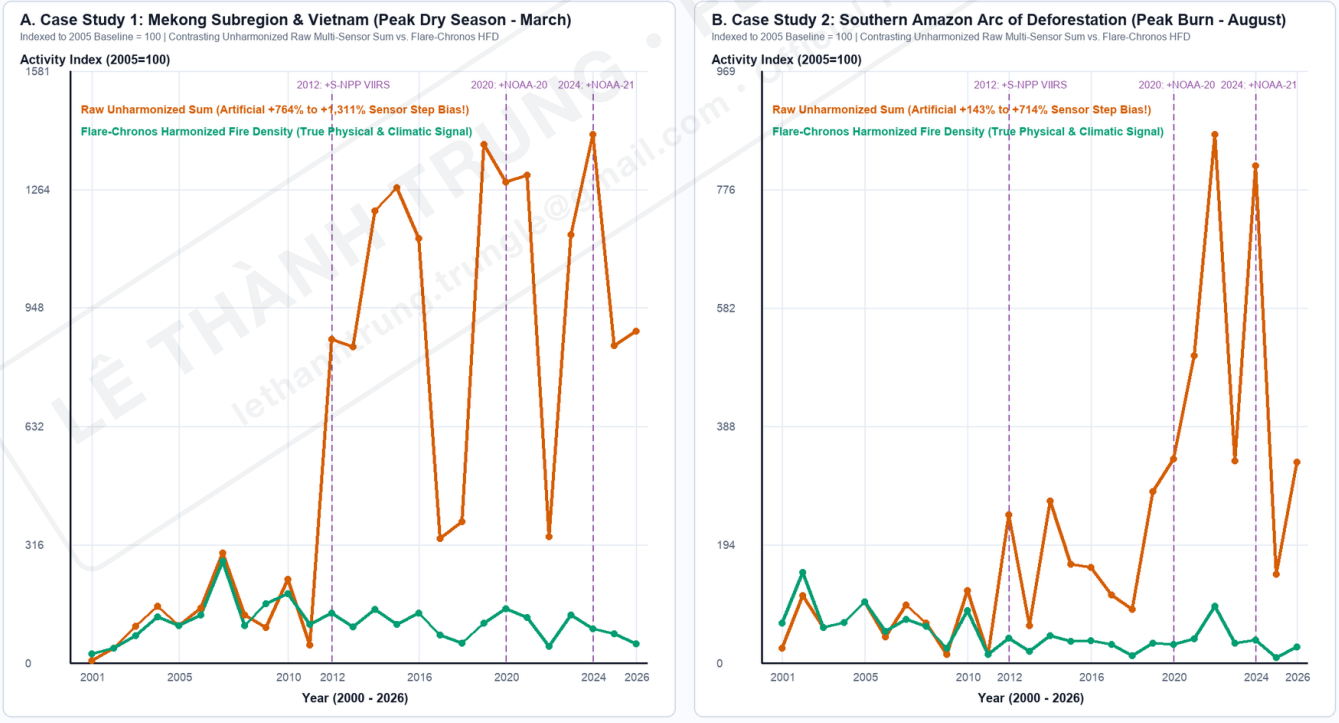



**Figure 4 - Regime Shift Elimination**

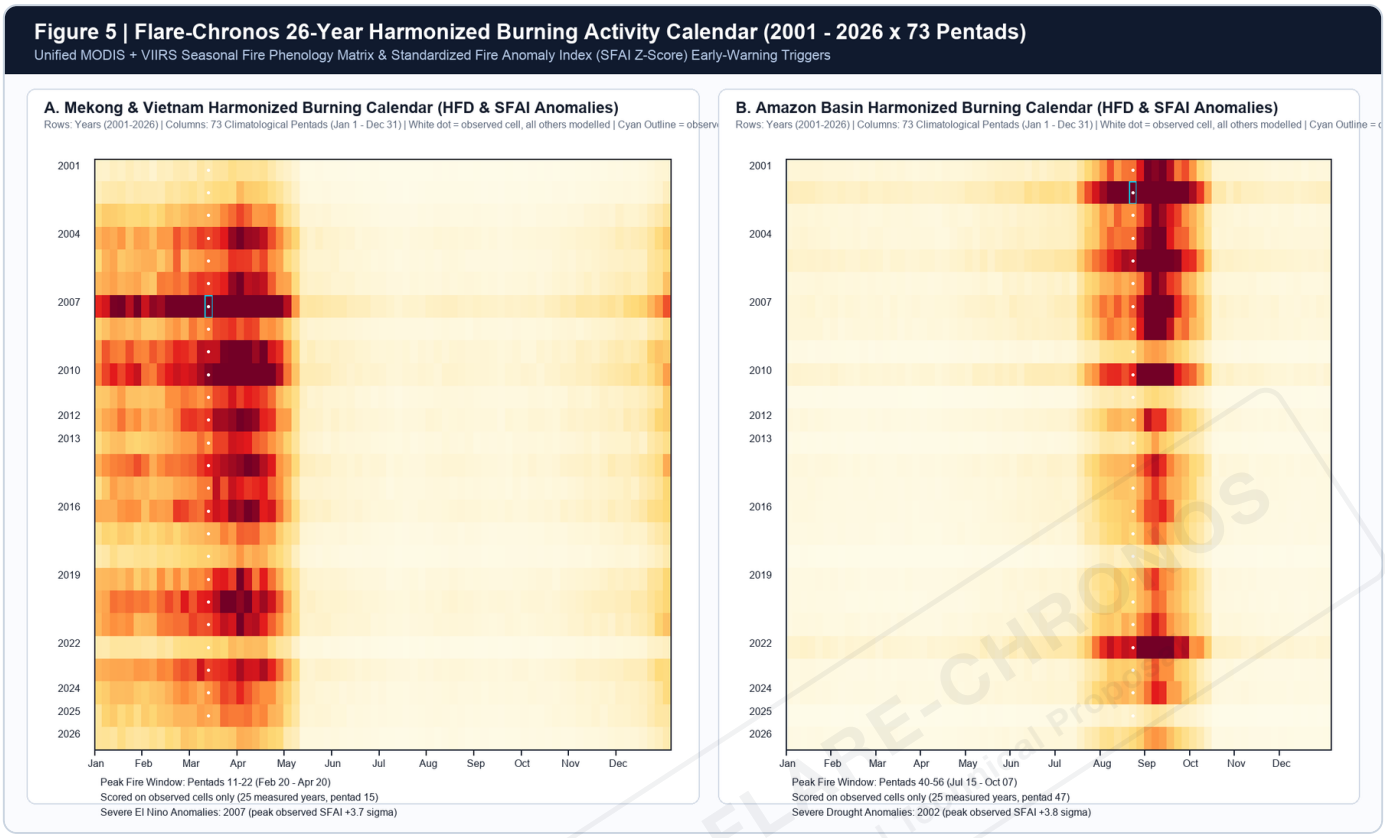
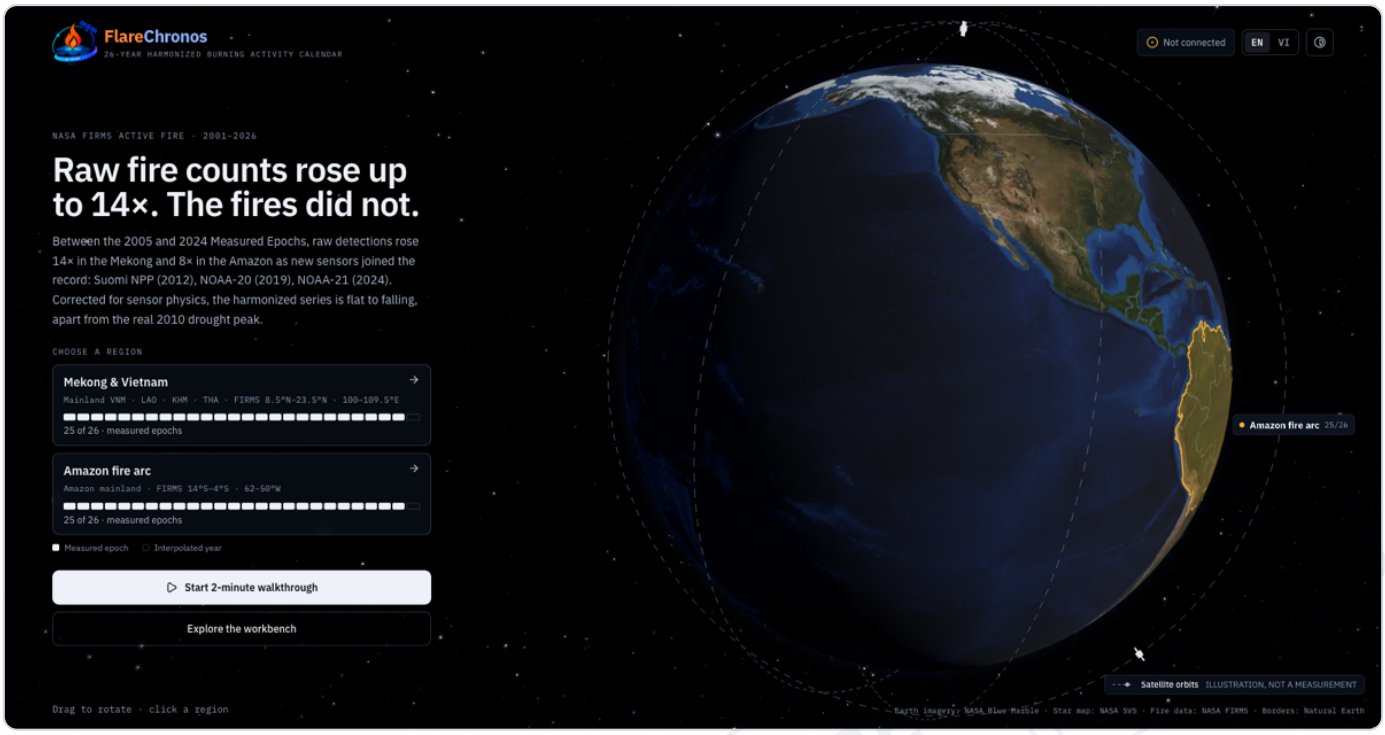



Figure 5 - Harmonized Burning Activity Calendar

## 7. Interactive Workbench, Dual-System Cognitive AI Harness & 2027+ Roadmap

### 7.1. Interactive Analyst Workbench ( `apps/workbench` ) & Dual-System AI Harness ( `apps/ai-service` )

1. **Interactive Web Surface ( `apps/workbench` , port `5900` )**:
   - Built with **Next.js 16 + React 19**, featuring a **Three.js 3D Region Globe** with camera fly-to transitions into a pre-projected `d3-geo` **1000×730 equal-area SVG map** ( `1–64×` smooth pan/zoom) with **Uber H3 Resolution 7 & 8 screen-space hexagonal binning**, elliptical scan footprints (**Δ S × Δ T**), the **26-Year × 73-Pentad Burning Activity Calendar**, the `/methodology` inspection view, the `/benchmark` 15-Use-Case evaluation matrix, and the `/review` human-in-the-loop sign-off workbench.
   - Wired directly to the live **FastAPI ASGI analysis server ( `apps/ai-service` , port `5901` )** with zero mock fixtures or local simulators (ADR-0016).




Interactive 3D Region Globe & Analyst Workbench Preview

1. **Dual-System Cognitive AI Harness (`apps/ai-service`, port `5901`)**:
   - **System 1 Reflexive Layer**: Assigns the request to `1` of `15` **Use Cases (`5` System 1 + `10` System 2)** using calibrated probabilities (`logprob` from the primary decision model, with automatic fallback to `OpenAIStructuredClassifierModel` `self_reported` confidence and deterministic regex extraction). Executes System 1 queries in `~1.4 ms` via a single Atomic Tool with zero LLM prose generation.
   - **System 2 Deliberative Layer**: Orchestrates multi-step scientific diagnosis across `18` **registered model-facing** `@tool` **functions (`9` Atomic + `9` Domain Diagnostic,** `19` `@tool` **functions total including the internal CSV harmonizer)**.
   - **Zero Hallucination & Human Sign-Off**: Every scalar in a briefing is injected from the `VerifiedFactsLedger` (`{{cite:FACT_ID}}`) and audited by `NumericGroundingVerifier` (`15/15` pass in both deterministic and live Gemini benchmarks, `0` unverified numbers). Operational dispatches (`UC-S1.4`, `UC-S2.6`, `UC-S2.9`, `UC-S2.10`) pause for an identity-bound JWT approval recorded in an append-only SQLite `audit` table.

### 7.2. Multi-Sensor Extension Roadmap (`2027+`)

1. **Geostationary 10-Minute Diurnal Fire Cycle (`GOES-16/19 ABI`, `Himawari-9 AHI`, `MTG-FCI`)**: Extend the scan-angle observation operator to geostationary imagers to reconstruct continuous 10-minute diurnal fire radiative energy (**FRE**) curves.
2. **30-Meter Sub-Field Calibration via `LANDSAT_NRT` (`Landsat 8/9/Next`)**: Anchor the low-FRP tail (**< 0.5 MW**) of agricultural residue burns using NASA FIRMS `LANDSAT_NRT` 30 m thermal detections.
3. **All-Weather L/S-Band SAR Burn Severity Fusion (`NISAR GCOV / GUNW`)**: Fuse harmonized thermal episodes with **NISAR** radar coherence-loss maps to verify canopy consumption beneath persistent tropical smoke plumes.

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# 8. Alignment With NASA Global Judging Criteria & Reproducibility Guide

## 8.1. Alignment With the Five Official NASA Space Apps Judging Criteria

| NASA Judging Criterion | Core Evaluation Focus | FLARE-CHRONOS Verifiable Contribution |
|---|---|---|
| **1. Impact** | Quantity and quality of real-world problem solving for science & society. | Eliminates the `+1,311.2%` **(** `+1.311,2%` **)** multi-sensor step-jump mirage across 26 years ( `2000–2026` ), preventing false climate policy conclusions and delivering actionable 5-day Pentad early warnings ( `SFAI Z-Score` ) for Mekong and Amazon forestry agencies. |
| **2. Creativity** | Novelty of mathematical, architectural, and visual approach. | Combines 13-point spherical elliptical footprint quadrature onto dual-resolution **Uber H3** hexagons, a two-stage physics + quantile-mapping FRP operator, a `26 × 73` Pentad Burning Calendar, and a zero-hallucination **Dual-System LangGraph Harness** with identity-bound human sign-off. |
| **3. Validity** | Scientific rigor, empirical validation on real NASA data, and automated testing. | Validated on `1,609,182` **real NASA FIRMS detections (** `134` **cached CSV slices)** and **N = 20,786 co-incident Aqua–SNPP pairs**, cutting $D_{KS}$ by `69.4%` ( `0.1153 → 0.0353` ) and raising $R^2$ from `0.7839` to `0.8048` . Guarded by `182` **offline pytest tests +** `6` **live tests across** `22` **test files**. |
| **4. Relevance** | Direct alignment with the *Harmonization of MODIS and VIIRS Hot Spots* challenge. | Solves all five cross-sensor discontinuities ( `Scan-Angle Footprint` , `FRP Detection Limit` , `Overpass Redundancy` , `SZA Day/Night Drift` , and `SAA / Static Noise` ) across both `Standard Processing` and `NRT` streams as MODIS and S-NPP retire in `2026` . |
| **5. Presentation** | Clarity of visual communication, interactive demo, and reproducibility. | Ships 5 publication-grade scientific figures ( `300 DPI` ), 3 English A4 technical PDFs ( `PROPOSAL.pdf` , `REPORT.pdf` , `MODEL_FLOW.pdf` ), and a live bilingual EN/VI **Next.js 16 + Three.js 3D Globe + H3 Workbench**. |

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### 8.2. End-to-End Reproducibility Commands

```
# 1. Run the full offline test suite (182 tests across 22 test files; 0 network calls)
uv run python -m pytest -q

# 2. Verify documentation, canonical claims, and PDF/MD synchronization
python3 scripts/check-docs.py --scan

# 3. Re-execute the empirical harmonization pipeline (451,488 benchmark records) & render 5 figures
uv run python -m benchmark_cli.run_experiments

# 4. Run the Dual-System AI Service (:5901) and Next.js 16 Workbench (:5900)
AI_SERVICE_AUTH=disabled uv run python -m ai_service.serve
cd apps/workbench && bun run dev
```

## 9. Appendix A — Full 25-Year Measured Epoch Tables (2001–2025)

Every row below is a **Measured Epoch** ( `25` Measured Epochs per region, `2001–2025` ), computed directly from real NASA FIRMS CSV slices ( `Mekong: March 15–19` ; `Amazon: August 20–24` ). Year `2026` is the single **Interpolated Year** because its annual window is still in progress. Both English ( `9,005.1` ) and Vietnamese ( `9.005,1` ) decimal notations are shown for cross-document audit traceability.

### 9.1. Mekong Subregion & Vietnam ( `25` Measured Epochs, `2001–2025` , `BBOX: 100.0°E–109.5°E, 8.5°N–23.5°N` )

| Year | Active Sensor Streams | Raw Detections | Fire Episodes | Effort | Harmonized `VIIRS-eq` Episodes (EN / VI) | Raw Index ( `2005=100` ) | Harmonized Index ( `2005=100` ) |
|---|---|---|---|---|---|---|---|
| **2001** | `MODIS_SP` (Terra) | 238 | 179 | 0.2785 | **2,191.7** ( `2.191,7` ) | 6.1 | **24.3** |
| **2002** | `MODIS_SP` (Terra+Aqua) | 1,552 | 1,054 | 1.0290 | **3,492.8** ( `3.492,8` ) | 39.7 | **38.8** |
| **2003** | `MODIS_SP` (Terra+Aqua) | 3,828 | 2,394 | 1.2363 | **6,603.2** ( `6.603,2` ) | 97.8 | **73.3** |
| **2004** | `MODIS_SP` (Terra+Aqua) | 5,893 | 3,614 | 1.1065 | **11,137.6** ( `11.137,6` ) | 150.6 | **123.7** |
| **2005** | `MODIS_SP` (Terra+Aqua) | 3,914 | 2,415 | 0.9145 | **9,005.1** ( `9.005,1` ) | 100.0 | **100.0** |
| **2006** | `MODIS_SP` (Terra+Aqua) | 5,727 | 3,294 | 0.9741 | **11,531.2** ( `11.531,2` ) | 146.3 | **128.1** |
| **2007** | `MODIS_SP` (Terra+Aqua) | 11,515 | 6,748 | 0.9413 | **24,445.6** ( `24.445,6` ) | 294.2 | **271.5** |
| **2008** | `MODIS_SP` (Terra+Aqua) | 4,986 | 3,148 | 1.2003 | **8,943.3** ( `8.943,3` ) | 127.4 | **99.3** |
| **2009** | `MODIS_SP` (Terra+Aqua) | 3,731 | 2,203 | 0.5273 | **14,246.6** ( `14.246,6` ) | 95.3 | **158.2** |
| **2010** | `MODIS_SP` (Terra+Aqua) | 8,726 | 4,990 | 1.0221 | **16,648.0** ( `16.648,0` ) | 222.9 | **184.9** |
| **2011** | `MODIS_SP` (Terra+Aqua) | 1,875 | 1,158 | 0.4278 | **9,230.4** ( `9.230,4` ) | 47.9 | **102.5** |

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| Year | Active Sensor Streams | Raw Detections | Fire Episodes | Effort | Harmonized VIIRS-eq Episodes (EN / VI) | Raw Index (2005=100) | Harmonized Index (2005=100) |
|---|---|---|---|---|---|---|---|
| **2012** | MODIS_SP + SNPP_SP | 33,828 | 16,845 | 1.4019 | **12,015.8** (12.015,8) | 864.3 | **133.4** |
| **2013** | MODIS_SP + SNPP_SP | 33,019 | 16,614 | 1.9300 | **8,608.3** (8.608,3) | 843.6 | **95.6** |
| **2014** | MODIS_SP + SNPP_SP | 47,234 | 23,921 | 1.8657 | **12,821.5** (12.821,5) | 1,206.8 | **142.4** |
| **2015** | MODIS_SP + SNPP_SP | 49,654 | 24,899 | 2.6747 | **9,309.1** (9.309,1) | 1,268.6 | **103.4** |
| **2016** | MODIS_SP + SNPP_SP | 44,336 | 23,500 | 1.9621 | **11,977.0** (11.977,0) | 1,132.8 | **133.0** |
| **2017** | MODIS_SP + SNPP_SP | 13,011 | 7,334 | 1.1006 | **6,663.6** (6.663,6) | 332.4 | **74.0** |
| **2018** | MODIS_SP + SNPP_SP | 14,742 | 9,123 | 1.9079 | **4,781.7** (4.781,7) | 376.6 | **53.1** |
| **2019** | MODIS_SP + SNPP_SP + NOAA20_SP | 54,171 | 24,647 | 2.0765 | **9,572.2** (9.572,2) | 1,384.0 | **106.3** |
| **2020** | MODIS_SP + SNPP_SP + NOAA20_SP | 50,218 | 22,409 | 1.3849 | **13,049.2** (13.049,2) | 1,283.0 | **144.9** |
| **2021** | MODIS_SP + SNPP_SP + NOAA20_SP | 50,988 | 23,642 | 1.7364 | **10,980.3** (10.980,3) | 1,302.7 | **121.9** |
| **2022** | MODIS_SP + SNPP_SP + NOAA20_SP | 13,165 | 7,063 | 1.4095 | **4,041.1** (4.041,1) | 336.4 | **44.9** |
| **2023** | MODIS_SP + SNPP_SP + NOAA20_SP | 44,775 | 22,164 | 1.5596 | **11,460.8** (11.460,8) | 1,144.0 | **127.3** |
| **2024** | MODIS_SP + SNPP_SP + NOAA20_SP + NOAA21_NRT | 55,235 | 24,902 | 2.0645 | **8,150.0** (8.150,0) | 1,411.2 | **90.5** |
| **2025** | MODIS_SP + SNPP_SP + NOAA20_SP + NOAA21_NRT | 33,171 | 15,457 | 1.4820 | **7,047.2** (7.047,2) | 847.5 | **78.3** |

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### 9.2. Southern Amazon Basin, Brazil ( `25` Measured Epochs, `2001–2025`, `BBOX: 62.0°W–50.0°W, 14.0°S–4.0°S` )

| Year | Active Sensor Streams | Raw Detections | Fire Episodes | Effort | Harmonized `VIIRS-eq` Episodes (EN / VI) | Raw Index ( `2005=100` ) | Harmonized Index ( `2005=100` ) |
|---|---|---|---|---|---|---|---|
| **2001** | `MODIS_SP` (Terra) | 3,364 | 1,800 | 0.3878 | **8,586.9** ( `8.586,9` ) | 24.4 | **65.5** |
| **2002** | `MODIS_SP` (Terra+Aqua) | 15,280 | 7,573 | 0.7202 | **19,453.0** ( `19.453,0` ) | 110.7 | **148.3** |
| **2003** | `MODIS_SP` (Terra+Aqua) | 8,040 | 4,034 | 0.9869 | **7,562.0** ( `7.562,0` ) | 58.2 | **57.7** |
| **2004** | `MODIS_SP` (Terra+Aqua) | 9,116 | 4,513 | 0.9612 | **8,686.1** ( `8.686,1` ) | 66.0 | **66.2** |
| **2005** | `MODIS_SP` (Terra+Aqua) | 13,804 | 6,955 | 0.9810 | **13,116.0** ( `13.116,0` ) | 100.0 | **100.0** |
| **2006** | `MODIS_SP` (Terra+Aqua) | 5,932 | 3,084 | 0.8337 | **6,843.5** ( `6.843,5` ) | 43.0 | **52.2** |
| **2007** | `MODIS_SP` (Terra+Aqua) | 13,033 | 6,369 | 1.2602 | **9,349.8** ( `9.349,8` ) | 94.4 | **71.3** |
| **2008** | `MODIS_SP` (Terra+Aqua) | 9,049 | 4,541 | 1.0583 | **7,938.1** ( `7.938,1` ) | 65.6 | **60.5** |
| **2009** | `MODIS_SP` (Terra+Aqua) | 1,907 | 1,065 | 0.6285 | **3,134.8** ( `3.134,8` ) | 13.8 | **23.9** |
| **2010** | `MODIS_SP` (Terra+Aqua) | 16,304 | 7,635 | 1.2589 | **11,219.9** ( `11.219,9` ) | 118.1 | **85.5** |
| **2011** | `MODIS_SP` (Terra+Aqua) | 1,928 | 1,062 | 1.0490 | **1,872.9** ( `1.872,9` ) | 14.0 | **14.3** |
| **2012** | `MODIS_SP` + `SNPP_SP` | 33,484 | 9,351 | 1.7705 | **5,281.6** ( `5.281,6` ) | 242.6 | **40.3** |
| **2013** | `MODIS_SP` + `SNPP_SP` | 8,494 | 2,678 | 1.0858 | **2,466.4** ( `2.466,4` ) | 61.5 | **18.8** |
| **2014** | `MODIS_SP` + `SNPP_SP` | 36,598 | 9,791 | 1.6831 | **5,817.2** ( `5.817,2` ) | 265.1 | **44.4** |
| **2015** | `MODIS_SP` + `SNPP_SP` | 22,285 | 7,010 | 1.4920 | **4,698.4** ( `4.698,4` ) | 161.4 | **35.8** |
| **2016** | `MODIS_SP` + `SNPP_SP` | 21,664 | 6,217 | 1.3128 | **4,735.7** ( `4.735,7` ) | 156.9 | **36.1** |
| **2017** | `MODIS_SP` + `SNPP_SP` | 15,401 | 4,974 | 1.2574 | **3,955.8** ( `3.955,8` ) | 111.6 | **30.2** |
| **2018** | `MODIS_SP` + `SNPP_SP` + `NOAA20_SP` | 12,106 | 3,121 | 1.6624 | **1,514.0** ( `1.514,0` ) | 87.7 | **11.5** |
| **2019** | `MODIS_SP` + `SNPP_SP` + `NOAA20_SP` | 38,751 | 8,764 | 1.6438 | **4,299.6** ( `4.299,6` ) | 280.7 | **32.8** |

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| Year | Active Sensor Streams | Raw Detections | Fire Episodes | Effort | Harmonized VIIRS-eq Episodes (EN / VI) | Raw Index (2005=100) | Harmonized Index (2005=100) |
|---|---|---|---|---|---|---|---|
| **2020** | MODIS_SP + SNPP_SP + NOAA20_SP | 46,112 | 10,446 | 2.1350 | **3,945.8** (3.945,8) | 334.0 | **30.1** |
| **2021** | MODIS_SP + SNPP_SP + NOAA20_SP | 69,450 | 13,537 | 2.1220 | **5,144.6** (5.144,6) | 503.1 | **39.2** |
| **2022** | MODIS_SP + SNPP_SP + NOAA20_SP | 119,492 | 23,196 | 1.5376 | **12,166.0** (12.166,0) | 865.6 | **92.8** |
| **2023** | MODIS_SP + SNPP_SP + NOAA20_SP | 45,718 | 9,686 | 1.8570 | **4,206.4** (4.206,4) | 331.2 | **32.1** |
| **2024** | MODIS_SP + SNPP_SP + NOAA20_SP + NOAA21_NRT | 112,405 | 19,542 | 2.6716 | **4,942.4** (4.942,4) | 814.3 | **37.7** |
| **2025** | MODIS_SP + SNPP_SP + NOAA20_SP + NOAA21_NRT | 20,029 | 3,780 | 2.2721 | **1,124.1** (1.124,1) | 145.1 | **8.6** |

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